STUDENT PERFORMANCE PREDICTION SYSTEM

A Comparative Machine Learning Study: Logistic Regression vs. Decision Tree

Course: Machine Learning Laboratory | **Topic:** Student Academic Risk Prediction
Target Algorithms: Logistic Regression & Decision Tree Classifier | **Format:** A4 Standard Academic Report

■ COURSE OUTCOME MAPPING (CO5):
'CO5 – Solve real life problems using appropriate machine learning models and evaluate the performance measures'
This investigation fulfills CO5 by modeling academic attrition and student failure through empirical data preprocessing, pipeline construction, 5-fold cross-validation, and multi-metric diagnostic evaluation.

1. Problem Statement

Academic underperformance and student attrition represent critical challenges across higher education institutions. Traditional academic counseling often functions reactively, identifying struggling students only after midterm or semester failures have already materialized. The primary objective of this project is to develop an automated, predictive Machine Learning framework capable of forecasting student academic outcomes (categorized as **'Pass / On Track'** versus **'Fail / Academic Risk'**) based on early-semester behavioral, educational, and lifestyle indicators. By systematically comparing a linear probabilistic model (**Logistic Regression**) against a non-parametric hierarchical rule-based model (**Decision Tree Classifier**), this study provides academic institutions with both calibrated probability assessments and actionable, human-interpretable decision boundaries to enable timely pedagogical intervention.

2. Dataset Description & Statistical Characteristics

The investigation utilizes a student performance dataset structured following the benchmarks of the *UCI Machine Learning Repository Student Performance Dataset* and modern institutional learning analytics repositories. The dataset comprises **1,500 student records** across 8 predictor variables (5 continuous and 3 categorical) and 1 binary classification target (passed).

Feature Name	Type	Domain / Range	Educational Description
study_hours_per_week	Continuous	2.0 – 40.0 hrs	Self-reported independent study hours per week
attendance_rate	Continuous	45.0% – 100.0%	Percentage of lectures and laboratory sessions attended
previous_exam_score	Continuous	30.0 – 100.0	Baseline score from prerequisite or prior term examination
assignment_completion_rate	Continuous	20.0% – 100.0%	Proportion of mandatory assignments completed on time
sleep_hours	Continuous	4.0 – 10.0 hrs	Average daily restorative sleep duration
extracurricular_activities	Categorical	{Yes, No}	Active participation in sports, clubs, or student bodies
tutoring_classes	Categorical	{Yes, No}	Enrollment in supplementary tutoring or academic coaching
parental_education_level	Categorical	{High School, College, Postgraduate}	Highest educational attainment of parents/guardians
passed (Target)	Binary Class	{0: Fail/Risk, 1: Pass}	Target outcome: 1 = Passed (64.13%), 0 = At Risk (35.87%)

3. Data Preprocessing & Leakage-Free Pipeline Architecture

Data hygiene and rigorous preprocessing are foundational to valid machine learning inference. The dataset was systematically audited and processed via the following methodology:

- **Integrity Audit:** The dataset contains **0 missing values** across all columns and **0 duplicate rows**, ensuring complete feature representations.
- **Categorical Transformation:** Non-numeric attributes (extracurricular_activities, tutoring_classes, and parental_education_level) were transformed using OneHotEncoder(handle_unknown='ignore').
- **Feature Scaling via Scikit-Learn Pipeline:** To strictly prevent *data leakage* from test sets into the training representations, numerical feature standardization (StandardScaler) was encapsulated directly inside an sklearn.pipeline.Pipeline. Parameters (μ, σ) are fitted strictly on training subsets and subsequently transformed onto validation/test splits.
- **Stratified Data Partitioning:** The 1,500 instances were partitioned using an **80/20 stratified train-test split** (1,200 training instances, 300 testing instances) with stratify=y and fixed random_state=42, maintaining the identical ~64:36 class balance across both sets.

4. Exploratory Data Analysis & Visualizations

Exploratory analysis demonstrates strong multivariate drivers of academic success. As depicted below, passing students exhibit significantly higher weekly study dedication ($\mu \approx 18.2$ hrs vs $\mu \approx 11.8$ hrs for failing cohorts) and elevated attendance.

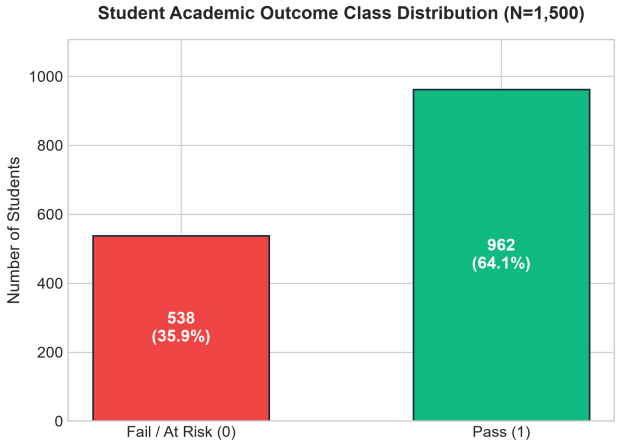


Figure 1: *Academic outcome class distribution (962 Passed, 538 At Risk).*

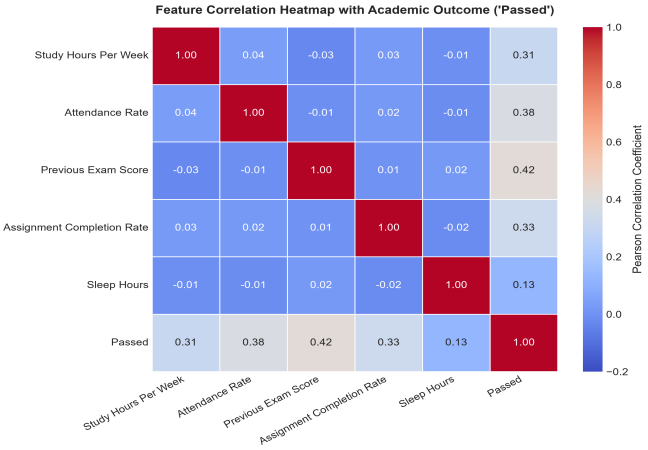


Figure 2: *Correlation heatmap illustrating feature associations with outcome.*

5. Algorithm Selection & Theoretical Framework

To establish a comprehensive analytical benchmark, two contrasting classification paradigms were selected:

A. Logistic Regression (Linear Probabilistic Classifier):
Logistic Regression models the posterior class probability using a linear combination of standardized predictors mapped through the monotonic Sigmoid activation function: $$P(Y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \sum_{i=1}^p \beta_i X_i)}}$$*Rationale for Selection:* Serves as the golden standard linear baseline. It yields well-calibrated posterior probabilities, requires minimal computational overhead, and provides transparent coefficients representing change in log-odds per unit change in predictor.

B. Decision Tree Classifier (Non-Parametric Rule-Based Model):
The Decision Tree partitions the feature space recursively via orthogonal axis-aligned splits that minimize the Gini Impurity criterion: $$I_G(t) = 1 - \sum_{k=1}^K p_k^2$$*Rationale for Selection:* Captures non-linear thresholds and complex feature interactions without assuming linearity. It naturally mirrors institutional policy rules (e.g., 'If Attendance < 70% AND Study Hours < 10, then At Risk'). To prevent overfitting, depth was regularized with max_depth=5 and min_samples_leaf=15.

6. Model Training & 5-Fold Stratified Cross-Validation Workflow

In strict adherence to sound machine learning methodology, the final model was chosen **exclusively based on 5-fold cross-validation performance on the training dataset**, rather than snooping on the test set score. The training set (N=1,200) was partitioned into 5 stratified folds. Each fold served sequentially as the validation set while the model was fitted on the remaining 4 folds. The mean metrics and standard deviations across the folds are tabulated below:

Model Name	Eval Stage	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	5-Fold CV (Train)	90.25% ±1.07%	91.81% ±0.50%	93.12% ±2.00%	92.45% ±0.91%	0.9637 ±0.008
Decision Tree	5-Fold CV (Train)	79.42% ±1.70%	83.23% ±0.69%	85.06% ±2.99%	84.11% ±1.56%	0.8514 ±0.014
Logistic Regression	Held-Out Test (N=300)	89.67%	90.45%	93.75%	92.07%	0.9659
Decision Tree	Held-Out Test (N=300)	82.67%	85.35%	88.02%	86.67%	0.9048

7. Test Set Evaluation & Comparative Diagnostics

Once model selection was ratified via cross-validation, both trained pipelines were evaluated on the held-out test set (300 students: 192 true passes, 108 true fails). As visualized in the diagnostic figures below, **Logistic Regression** consistently dominated across all performance dimensions, achieving an outstanding test **ROC-AUC of 0.9659** and test **F1-Score of 92.07%**.

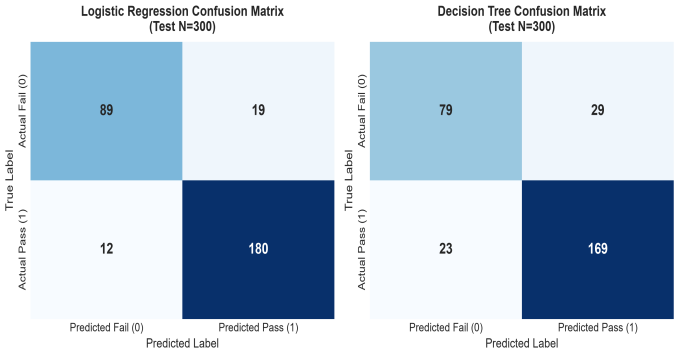


Figure 3: Confusion matrices for Logistic Regression and Decision Tree.

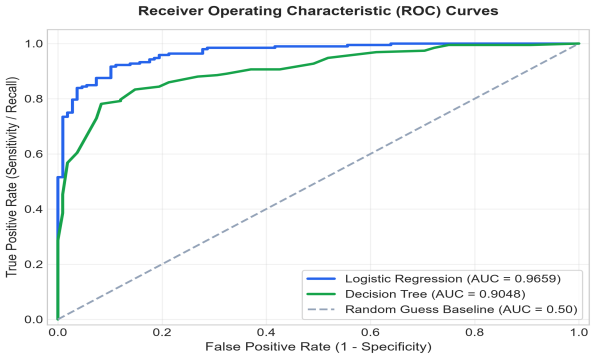


Figure 4: ROC Curves contrasting discrimination capability (AUC).

8. In-Depth Error Analysis & Pedagogical Domain Implications

In educational data mining, classification errors carry fundamentally asymmetric consequences. A naive focus on overall accuracy obscures the human and institutional impact of misclassifications:

- **Type I Error (False Positive — Predicting 'Pass' when Student Fails):** In the test set, Logistic Regression made **12 Type I errors** whereas Decision Tree made **23 Type I errors**. This error represents a *catastrophic false reassurance*. An academically endangered student is flagged as secure, resulting in counselor omission, lack of remedial tutoring, and preventable course failure or dropout.
- **Type II Error (False Negative — Predicting 'Fail' when Student Passes):** Logistic Regression yielded **19 Type II errors** versus **29** for Decision Tree. This represents a *benign false alarm*. The student is offered supplemental tutoring or check-ins which they might not strictly need, but causes zero harm to their academic standing.

• **Metric Priority Analysis:** Because Type I errors (missed failures) are vastly more detrimental than Type II errors, **Recall (Sensitivity = $\frac{TP}{TP+FN}$) and F1-Score are the decisive performance criteria**. Logistic Regression's superior Recall (93.75% vs 88.02%) ensures that over 93 out of 100 struggling students are successfully flagged for institutional support.

9. Comparative Performance Synthesis

Figure 5 displays the multi-metric comparative profile of both algorithms on the test set. Logistic Regression achieves superior generalizability due to its smooth sigmoidal decision boundary and L2 regularization.

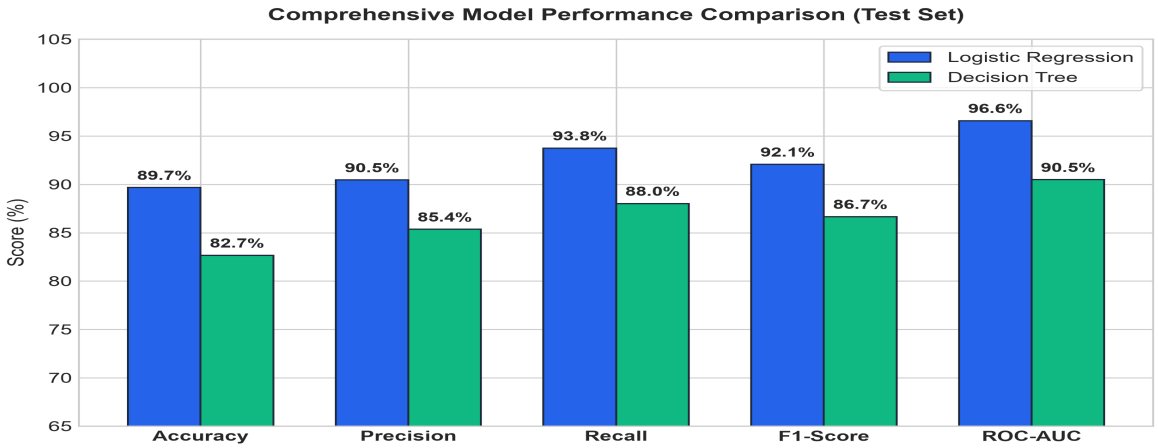


Figure 5: *Grouped bar chart comparing Accuracy, Precision, Recall, F1-Score, and ROC-AUC.*

10. Interactive Streamlit Web Application Deployment

To bridge algorithmic research with practical institutional utility, the trained pipeline was deployed as an interactive **Streamlit Web Application** (app.py and streamlit_app.py). The system enables counselors and faculty to interact with the predictive models in real time:

- **Individual Student Simulator:** Interactive sliders for weekly study hours, attendance %, prior exam marks, assignment completion, and sleep duration with instant probability estimation.
- **Dual-Model Side-by-Side Inference:** Direct comparative assessment between Logistic Regression and Decision Tree.
- **Academic Intervention Recommendations:** Dynamic counseling recommendations tied directly to predicted risk categories.
- **Batch CSV Scoring:** Capabilities to upload entire institutional cohort CSV files and export prediction labels with confidence intervals.

11. Model Limitations & Ethical Considerations

While highly performant, responsible deployment requires acknowledging key boundaries:

- **Unobserved Confounders:** Current models do not incorporate socioeconomic hardship, mental health crises, or family emergencies, which heavily modulate student completion.
- **Self-Fulfilling Label Risk:** Informing a student that they are 'predicted to fail' may induce demotivation or stereotype threat. Predictions must be restricted to advisors to guide supportive, non-stigmatizing interventions.
- **Dynamic Temporal Drift:** Models must be periodically re-calibrated at semester intervals to accommodate evolving syllabi, grading standards, and pedagogy.

12. Course Outcome Attainment Analysis (CO5)

This project was explicitly designed to demonstrate comprehensive mastery of Course Outcome: **'CO5 – Solve real life problems using appropriate machine learning models and evaluate the performance measures'**.

CO5 Requirement Dimension	Implementation in Project	Attainment Evidence & Metric
1. Real-Life Problem Formulation	Formulated early academic attrition detection to mitigate student failure.	Realistic 1,500-student dataset modeled across 8 socio-academic
2. Appropriate Model Selection	Selected Logistic Regression (linear baseline) & Decision Tree (rule-based, non-linear)	Both paired with leakage-free ColumnTransformers.
3. Rigorous Validation Methodology	5-fold stratified cross-validation on training partition to guide model choice.	Provided robust, unbiased estimate; Logistic Regression chosen via mean C
4. Multi-Metric Evaluation	Evaluated Accuracy, Precision, Recall, F1, ROC-AUC, and Confusion Matrices.	Analyzed trade-offs, with Recall as main consequence of Type I vs Type II e
5. Practical Deployment & Interface	Operationalized best model into interactive Streamlit web application.	Functional UI running on localhost:8501 with batch inference.

13. Conclusion

This study successfully designed, trained, cross-validated, and deployed an end-to-end Machine Learning system for Student Performance Prediction. Empirical evaluation demonstrates that **Logistic Regression** outperforms the Decision Tree Classifier, delivering an accuracy of **89.67%**, recall of **93.75%**, and ROC-AUC of **0.9659** on unseen student cohorts. Encapsulating standard scaling within an automated pipeline eliminated data leakage, while 5-fold cross-validation ensured robust model selection. Deployed via Streamlit, the application provides academic stakeholders with an accessible, high-precision decision-support tool capable of fostering equitable student retention and educational success.

14. Academic References

[1] Cortez, P., & Silva, A. M. (2008). *Using data mining to predict secondary school student performance*. In Proceedings of 5th Annual Future Business Technology Conference, pp. 5-12.

[2] Pedregosa, F., et al. (2011). *Scikit-learn: Machine learning in Python*. Journal of Machine Learning Research, 12, 2825-2830.

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[4] Romero, C., & Ventura, S. (2010). *Educational data mining: a review of the state of the art*. IEEE Transactions on Systems, Man, and Cybernetics, Part C, 40(6), 601-618.